Chapter #3

Uncovering the Dark Side of Fiat

3.0 Class Exercise: The Effects of Inflation: An Auction Activity
3.1 The Biggest Threats to Your Money: Inflation, Debasement, and Purchasing Power Loss
3.2 Debt: The Fine Line Between Help and Harm
3.3 The Fed and It's Partners: How the Government and Banks Control Money Supply
3.4 The Magic of Money Creation
- **3.4.1** The Time Value of Money and Its Role in Economic Growth
- **3.4.2** Saving Money In a Hard Time
- **3.4.3** Fractional Reserve Banking
- **3.4.4** Class Exercise: Fractional Reserve Banking

3.0 Class Exercise: The Effects of Inflation: An Auction Activity

Objective: To understand the concept of the **money supply** and how it affects the prices of goods and services in an economy.

Definitions:

- **Money Supply** is the total amount of money in circulation within an economy at a specific time. This includes:
    - Physical currency, such as coins and bills
    - Electronic money held in bank accounts

  - The money supply is an important concept in economics, as it can affect the overall health of an economy.
- **Auction** is a public sale in which goods or property are sold to the highest bidder.

Class Exercise. Follow the instructions below.

1. You will receive a random amount of monopoly money from the teacher. This represents the money supply in a society.

2. Write down the total money supply in the chart provided.

3. The teacher will auction a candy bar to the students. To win the candy bar, you will need to make the highest bid using your monopoly money. Record the winning bid next to the money supply.

4. The teacher will then add a significant amount of monopoly money to the total money supply. This represents an increase in the money supply in an economy. Later, you will learn how money supply is added or reduced in an economy.

5. The teacher will auction a second candy bar to the students using the same process as before. Record the winning bid next to the money supply on the chart.

6. The teacher will repeat the auction a third time.

Societies can often be unpredictable and unjust, exemplified by the simulation of a teacher randomly giving a significant amount of money to only a select few students. This mimics real-life situations where unequal distribution of resources and opportunities can occur, highlighting the inherent randomness and unfairness in many situations.

Round	Money Supply	Winning Bid

Questions. Based on what you learned from the exercise, answer the following questions.

1. How did the increase in the money supply affect the winning bids for the candy bars?

2. What is the relationship between the money supply and inflation?

3. How is the money supply relevant in the real world?

4. Can you think of any other factors that can affect the prices of goods and services?

3.1 The Biggest Threats to Your Money: Inflation, Debasement, and the Loss of Purchasing Power

The current global economic climate is challenging, which can make it difficult to save. One factor contributing to this is inflation, a phenomenon that occurs when the value of money decreases over time. This means that even if you save more dollars now, it may not have the same purchasing power in the future, meaning that more money will actually buy you fewer things. Recognizing economic conditions and their impact on your personal finances will help you make informed decisions about saving and spending.

Purchasing power is the amount of goods or services that can be bought with a given amount of money.

1956 2020 2056

Let us begin with a realistic scenario to explain each term.

Jaime is a college student who lives in a small apartment. He works part-time at a coffee shop to pay for his living expenses and tuition. As soon as he began living independently, Jaime became a pro at managing his own **ledger**.

At the beginning of the year, he budgeted $10,000 for his living expenses, including rent, food, and other necessities.

Inflation is an increase in the general level of prices of goods and services in an economy over a period of time. When the general price level rises, each unit of currency buys fewer goods and services; consequently, inflation reflects a **reduction in the purchasing power of money** – a loss of real value in the medium of exchange and unit of account within an economy.

A **ledger** is a detailed record of all of your monetary transactions. Whether it's money you're earning or spending, a ledger helps you keep track of it all.

These were his transactions for January:

Date	Description	Amount	Type	Balance
01/01/2023	**Starting Balance**			**$1,600.00**
01/01/2023	**Rent for January**	$800.00	**Debit**	**$800.00**
01/05/2023	**Groceries**	$100.00	Debit	**$700.00**
01/15/2023	**Part-time paycheck**	**$500.00**	**Credit**	**$1,200.00**
01/20/2023	**Gas for car**	$50.00	Debit	**$850.00**
01/30/2023	**Textbooks**	$150.00	Debit	**$650.00**

This ledger shows that Jaime's starting balance on his checking account on January 1st was $1,600, out of which he spent $800 to pay rent for the month. He then spent (a **debit**) $100.00 on groceries and received $500.00 (a **credit**) in pay from his part-time job, bringing his balance to $1200.00. He then spent money on gas and textbooks, bringing his balance down to $650.00 at the end of the month.

Twelve months later, while having lunch with his grandfather, Jaime notices that his budget is not stretching as far as it used to. He realizes that the prices of the goods and services he needs have increased significantly over the past year, and wondered why. Then he saw this image and could not believe his eyes.

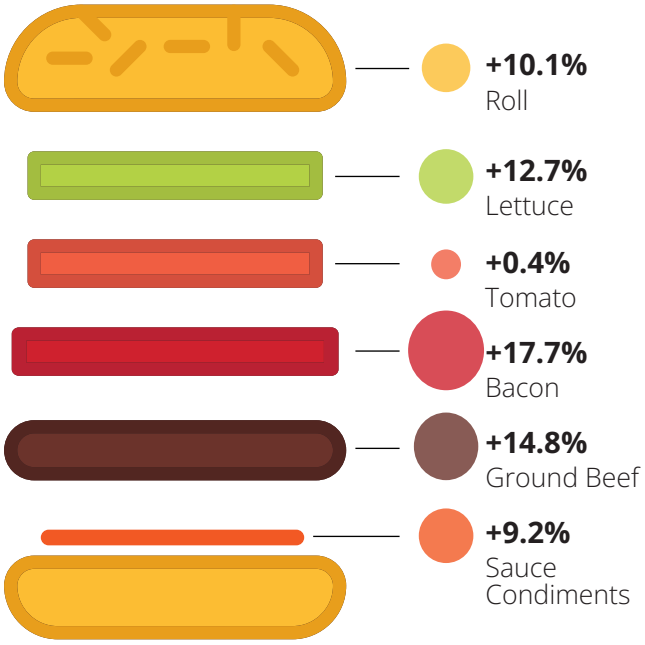
When he brought it up to his grandfather, he was told: "In 1956, I was just a young man starting out in the world. I remember that I used to earn $100 a month as a factory worker. It may not seem like much by today's standards, but it was a decent wage at the time. In fact, I was able to save up enough money to buy a small house in the suburbs."

As we can see, the cost of each item in the **basket** has increased, leading to an overall decrease in his purchasing power.

Thankfully, Jaime has mastered the use of a ledger, for it clearly showed him how his annual purchasing power has decreased.

How Inflation Changed the Price of a Hamburger

Year-over-year change in the price of selected ingredients of a hamburger (April 2021 – April 2022)



** Based on retail prices, urban consumers.*

Jaime: "What? That's crazy. I can't even imagine what my rent would have cost back then."

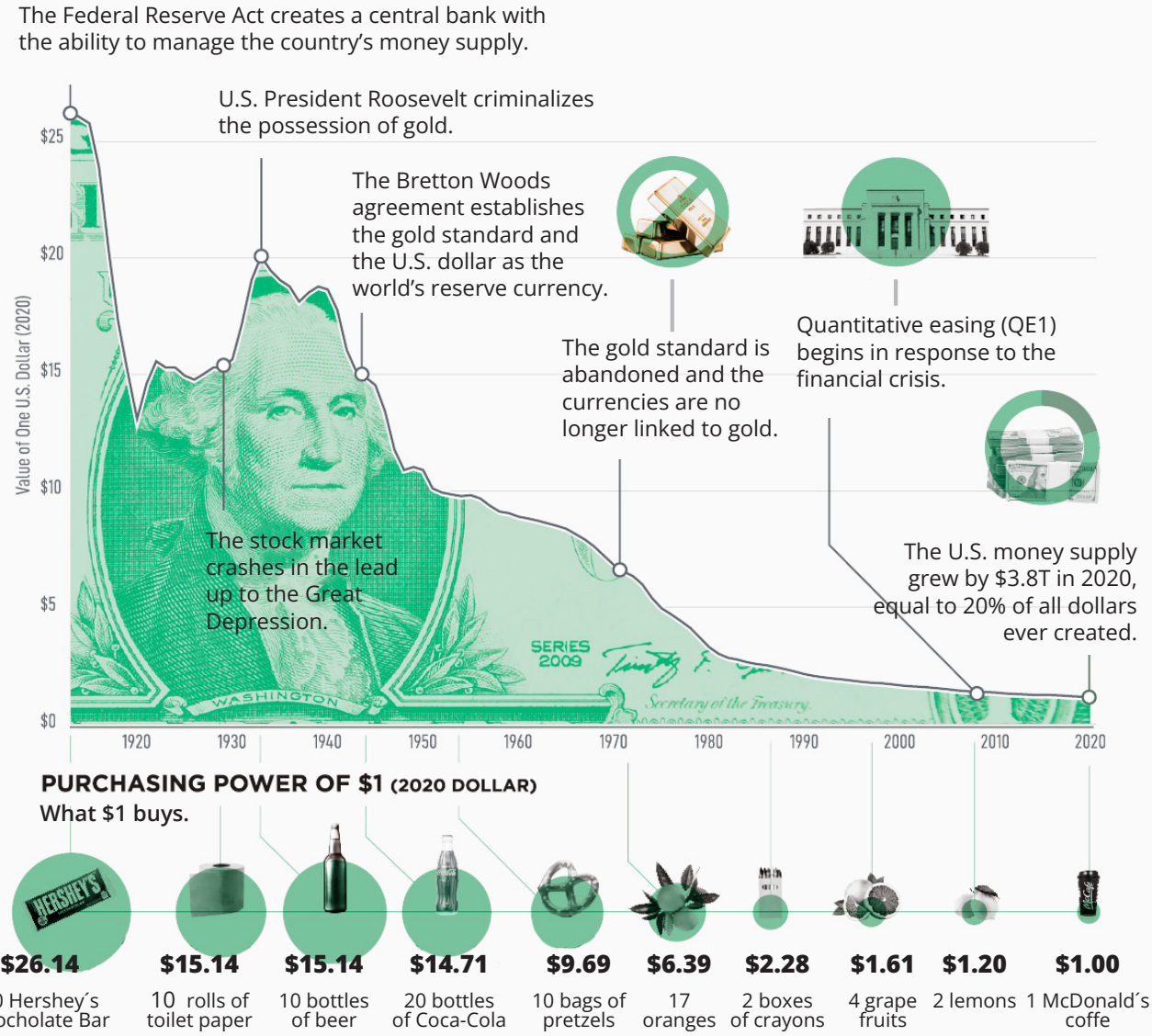
Grandfather: "Well, let me see. If we take inflation into account, $1USD would have bought me about 10 bags of pretzels back then."

Jaime: "Wow, that's really interesting, Grandpa. But how much would that be worth today?"

Grandfather: "Oh, things were much cheaper! A loaf of bread would only cost a few cents, and you could buy a gallon of gasoline for just a quarter. It is unbelievable how much the cost of living has gone up."

A Dollar's Worth
Purchasing Power of the U.S. Dollar

The purchasing power of the U.S. dollar has fallen sharply over the last century, due to rising inflation and money supply.



- Jaime needs to budget an additional $1,000 for the same basket of goods and services that he purchased the previous year.
    - This means that his purchasing power has decreased by $1,000, as he *now has to spend more money to buy the same goods and services*.
- The basket of goods and services includes rent for his apartment, groceries, and other necessities.
- The following table shows the cost of each item in the **basket** in the first year and the second year, as well as the percentage increase in price:

Item	Cost Year #1	Cost Year #2	% Increase
Rent	$4,000	$4,500	**12.5%**
Groceries	$2,000	$4,300	**15%**
Necessities	$4,000	$4,200	**5%**
Total	**$10,000**	**$11,000**	**10%**

Jaime earns more in a year than his grandfather ever did, but this also disincentivizes saving. It's more advantageous to spend money now since its value decreases. This hinders the ability to plan for the future. As shown in a previous graph (in Section 2.3), salary growth year over year in the United States has remained stagnant for the average citizen, meaning most people aren't receiving raises at the same rate as the decreasing value of their money, despite working harder.

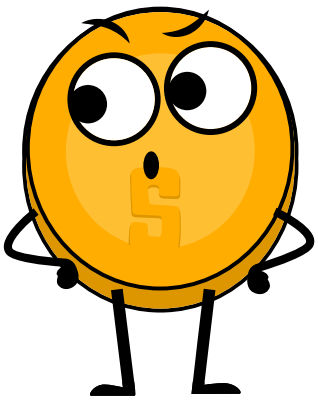
It could have been worse for Jaime. For example, Zimbabwe experienced hyperinflation in the late 2000's, when the country's economy was hit by a combination of political instability, economic mismanagement, and external factors such as drought and sanctions. As a result, the value of the Zimbabwean dollar (ZWD) plummeted, and the government was forced to print more money.

- The 100,000 ZWD bill was introduced in Zimbabwe in 2008. Because of hyperinflation, it was worth only a few US dollars at the time.
- Despite its high face value, the 100,000 ZWD bill was not enough to buy basic necessities like food or fuel, and people had to carry large bundles of cash to make everyday purchases.

After considering the significant price increases since the mid 50's in the US and the example of hyperinflation in Zimbabwe, it becomes clear that the impact of inflation on an individual's purchasing power can vary greatly depending on their location and the time period in which they lived.

Inflation tends to affect those who live in poor nations much more than it does those living in rich countries. This highlights the fact that it is often pure luck where and when an individual

So what exactly is inflation? Why is it so dangerous? Steve Forbes breaks it down.



is born, and that the circumstances of an individual's birth can have a significant impact on their quality of life and economic opportunities.

3.2 Debt: The Fine Line Between Help and Harm

Debt is a double-edged sword. It's true that borrowing money can provide a much-needed financial boost, whether it's for individuals making a big purchase, businesses investing in their growth, or governments funding important services. But borrowing too much can lead to financial ruin. When you can't pay the interest on your debts, it becomes harder to pay your bills and stay afloat. This is especially true when an entity takes on more debt to pay off existing debt, and gets caught in a vicious cycle known as a "debt spiral."

Debt is money that is owed by one person or organization to another. When you have debt, you are required to pay back the money you owe, usually with interest, by a certain date.

The debt crisis is a worldwide problem, including in the United States. Currently, the government is spending more money than it is earning. To pay its bills, it has been borrowing more and more money. However, this cycle of debt and higher borrowing costs can soon harm the government's credit rating. If the debt becomes too much, the government may have financial difficulties and potentially go bankrupt, just as many other countries have in the past.

The State of the World´s Government Debt

Debt to GDP Ratio 2021 (%)

- 200% and more
- 100% - 199.9%
- 50% - 99.9%
- 20% - 49.9%
- 10 – 19.9%
- Less than 10%

- Debt taken on by the government can have long-term effects on future generations.
- Printing more money to fund expenses can result in currency devaluation and a possible collapse of the monetary system.

But how can we measure the risk of a country taking on too much debt? One way is through the **debt-to-GDP ratio,** which shows the amount of a country's total debt as a percentage of its GDP.

- The **debt-to-GDP ratio** is a way to see if a country can pay its debts.
    - If the ratio is high, the country may have trouble paying its debts in the future.
    - If the ratio is low, the country may be able to pay its debts easily and be in good financial shape.
- It's important to remember that the debt-to-GDP ratio is only one part of understanding a country's money situation.

Gross Domestic Product (GDP) is a measure of the total value of goods and services produced within a country over a specific period of time, typically a year. It is often used as a measure of the size and health of an economy.

3.3 The Fed and Its Partners: How the Government and Banks Control the Money Supply

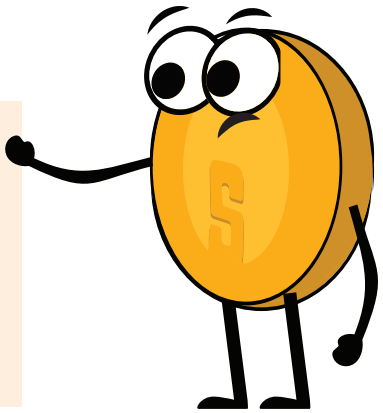
Have you ever stopped to consider where the trillions of dollars in stimulus funds that were distributed during the pandemic came from, and who gets to decide how much is given and to whom? The allocation of these funds has the power to greatly impact society and the economy, yet it often goes largely unexamined.

There are several tools that centralized governments can use to manage how much **money supply** there is at a specific moment in time.

- Central banks and governments can use monetary and fiscal policy tools to influence the money supply and the economy.

The Central Bank of the United States is called **The Federal Reserve**, or The Fed.

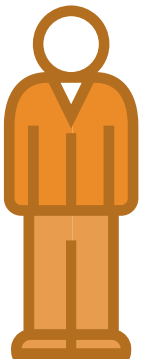
Governments may borrow money to stimulate the economy, but this can lead to inflation if they have to print more money to pay back the loans.



Uncovering the Dark Side of Fiat

Target Rates
Monetary Policy

Unemployment Below
6.5%



2% - 3%
Annual Increase in Gross Domestic Product

Core Inflation Rate between
2.0% - 2.5%

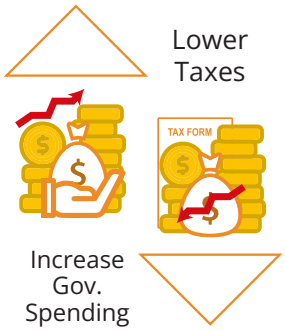
Expansionary
Fiscal Policy

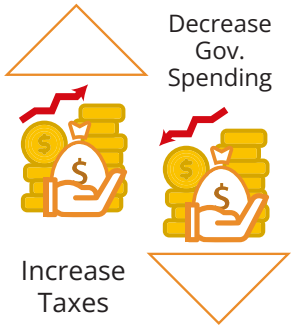
VS

Contractionary
Fiscal Policy

Aims to increase consumer spending and business investment to increase aggregate demand an economic growth.

Aims to increase consumer spending and business investment to increase aggregate demand an economic growth.





Policy/Tool	Description	Example
Monetary Policy	Monetary policy involves adjusting interest rates to control the amount of money in circulation.	**The Federal Reserve** raising interest rates to slow down inflation or can lower them to stimulate employment.
Fiscal Policy	Fiscal policy involves using spending and tax policies to influence the economy.	The government increasing spending on infrastructure projects to stimulate economic growth. It can also decrease taxes so people spend more.
Exchange Rate Policy	The use of a country's exchange rate (the value of its currency in relation to other currencies) to influence trade and the economy.	The Chinese government pegging the value of the yuan to the U.S. dollar to stabilize exchange rates
Supply Shock	A sudden and unexpected event that disrupts the supply of goods and services, leading to changes in prices and the money supply.	A natural disaster that destroys a significant portion of a country's agricultural production, leading to food shortages and price increases.
Price Controls	Government-imposed limits on the prices of goods and services to manage inflation or stabilize prices.	The government setting a maximum price for gasoline to prevent price gouging during a crisis.

3.4 The Magic of Money Creation

3.4.1 The Time Value of Money and Its Role in Economic Growth

Have you ever wondered why banks offer so many services to their customers? While it may seem like they are being generous, it's important to remember that banks are businesses, and their primary goal is to make a profit. But how do they make a profit if they are giving away money in loans?

In addition to earning interest on deposits, banks generate revenue in other ways, including:

1. Charging interest on loans they give out
2. Charging fees for services like ATM usage and account maintenance
3. Earning money through investments, like buying and selling securities or investing in real estate
4. Keeping a percentage of loans in reserve and investing or lending out the rest
5. Paying interest on deposits and charging fees on checking and savings accounts

By borrowing money at low interest rates and lending it out at higher rates, banks are able to turn a profit. They also generate income through fees and investment activities.

But why should this matter to you as an individual? Well, have you heard the phrase "a dollar today is worth more than a dollar tomorrow"? This concept is known as the **time value of money**, and it's all about the idea that money is worth more in the present than in the future. This is because ***money can be invested to earn interest and because money can lose value over time due to inflation***.

In other words, if you have money sitting in a savings account earning a low interest rate, it's not going to be worth as much in the future as it is today. On the other hand, if you invest your money in something that has the possibility of earning a higher return, you might come out ahead.

To ensure that your money retains its value over time, the goal of investing is to earn a return that is higher than the rate of inflation. This way, your money will be worth more in the future than it is today.

3.4.2 Saving Money in Hard Times

The current global economic situation, which was negatively affected by the pandemic, has brought about challenges such as high inflation and low interest rates on savings accounts. These conditions can make it tough to effectively save money, as inflation eats away at the value of currency over time. Even if you save today, you may end up with less purchasing power in the future.

But don't worry! There are still ways to save money and be financially secure. Here are a few ideas to try:

- **Make a budget**: A budget is a plan for how you will use your money. It can help you see where you are spending too much money and where you can save. Set aside some money each month for saving, and look for ways to cut back on your expenses.
- **Start investing**: Investing is a way to make your money grow over time. There are many types of investments to choose from, and you can find one that fits your budget and your level of risk.
- **Get creative**: There are many creative ways to save money. You can try cutting your own hair or bartering with others for goods and services. Be open to trying new things and looking for non-traditional solutions to your financial problems.

  - It is **generally acceptable** to **take on debt as long as the money is used to generate income and increase purchasing power in the future**. This is because borrowing money can allow an individual or business to make investments that increase their productivity and efficiency, ultimately leading to greater profits and financial stability.
  - For example, if a farmer takes out a loan to purchase new equipment that allows them to harvest their crops more quickly and efficiently, they may be able to generate more income and increase their purchasing power as a result. On the other hand, if the money is used to waste resources or make unproductive investments, it can lead to financial difficulties and wouldn't be a wise decision.

By taking charge of your finances and being flexible, you will be better able to weather the storm of a tough economy and come out on top.

The 50/30/20 Spending Plan

3.4.3 Fractional Reserve Banking

So far, we've talked about how central banks like the Federal Reserve manage the money supply, how banks make money for themselves, and a few strategies on how to save money, but we haven't yet discussed how new money is actually created and introduced into a society. It might seem like magic, but there is an interesting process behind it.

How does **new** money actually enter circulation and fuel economic growth? Unlike physical resources like food or water that can run out - money has no fixed limit! So how does it work?

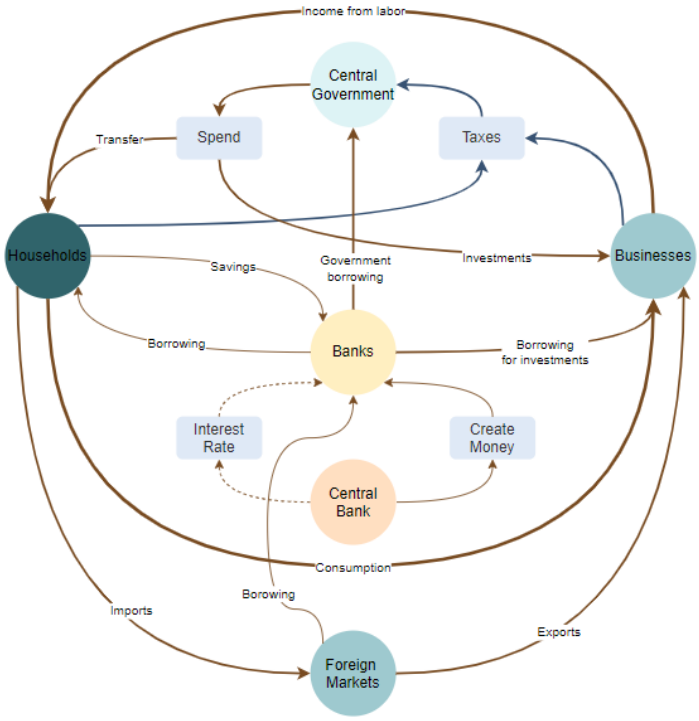
The **government** the **central bank**, and **private banks** all play a role in this process.

Here is a simplified version of how the Federal Reserve (Fed) can add $100 million into circulation:

1. The Fed determines that it wants to increase the money supply by $100 million. This decision is typically made based on the Fed's **monetary policy goals**, such as boosting economic growth or stabilizing prices.
2. The Fed instructs a large commercial bank to create a $100 million deposit in its account at the Fed. This deposit is created out of thin air and is not backed by any physical assets.

- When a commercial bank creates a deposit at the Fed, it is essentially borrowing money from the Fed. The Fed provides the bank with the funds for the deposit, and in return, the bank must pay interest on the loan and eventually pay the loan back.

3. The member bank then uses this new $100 million deposit to make loans to businesses or individuals, or to purchase securities such as government bonds.
4. The businesses or individuals who receive these loans can then use the money to make purchases, pay bills, or invest in other assets. This increases the overall supply of money in the economy.
5. As the money is circulated and spent, it eventually ends up in other banks, which can then use it to make their own loans and investments. This process continues until the $100 million has been fully injected into circulation.



Overall, the Fed's ability to add new money into circulation through the banking system helps to stimulate economic growth and achieve its monetary policy goals.

Banks actually create **new** money every time they lend to customers or make investments. What? Yes, you read that right. When a bank makes a loan, it creates money by adding *new funds* to the borrower's account for the loan amount. The borrower can then use this money to make purchases or pay bills, effectively increasing the overall supply of money in the economy. We will see how next.

3.4.4 Class Exercise: Fractional Reserve Banking

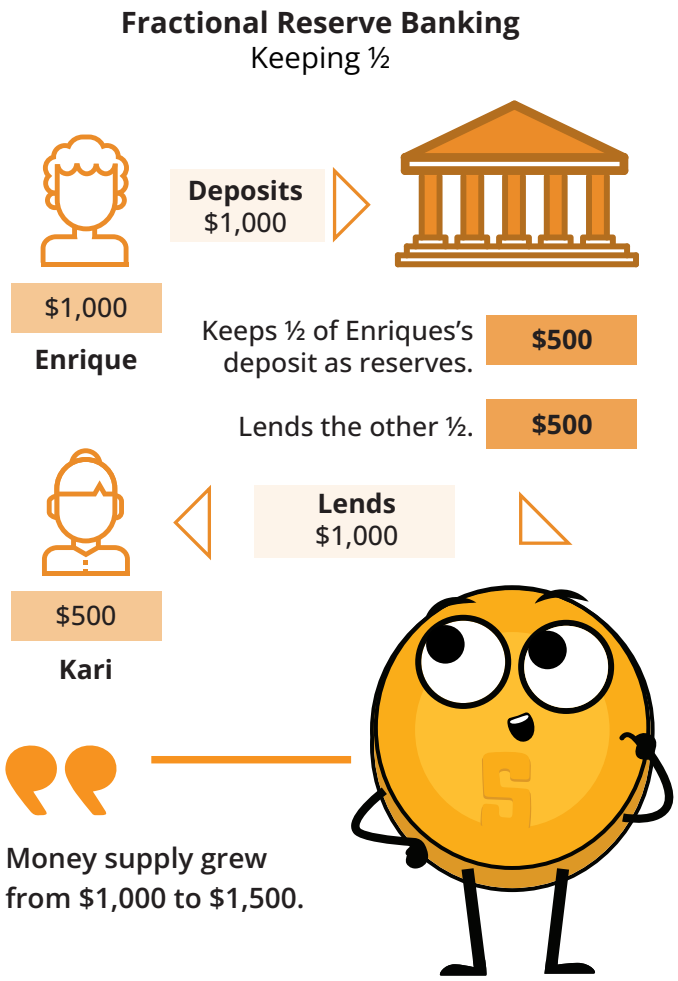
The process known as **fractional reserve banking** is where ***banks only hold a fraction of their deposits as reserves and lend out the rest.*** As long as they maintain a certain reserve ratio set by the central bank, banks can create more money than they have on hand. However, this ability to create new money can also carry the risk of excessive borrowing and financial instability if not managed carefully.

The **reserve ratio** is a rule that tells banks how much money they have to keep in their safe and how much they can lend out. It is **set by the central bank**, which is a special group of people in charge of making sure the economy is healthy.

In this activity, we will explore the concept of **fractional reserve banking** and how it can lead to the **debasement** of a currency; **inflation**; and a decrease in **purchasing power**.

- Let's say the total amount of money in the economy is $1000 and the reserve ratio is 50%. This means that for every $1000 in the economy, 50% of it must be kept in reserve by the bank.

- If Enrique deposits $1000 in the bank and later Kari comes to the bank for a loan, with the required ratio, the bank can keep half and lend half, so they would lend her $500. As a result, the total money supply would increase from $1000 to $1500.



The formula is: *Money Created = Total Amount of Money in the Economy ÷ Reserve Ratio*

Class Exercise. Using the formula above, we can calculate the amount of money created as follows:

- Money Created = **$1000 / 50% = $2000**

**Please note that this is a little oversimplified.*

We are going to model the creation of money in a small economy (which will consist of 6 participants; one will have the role of a bank). The **reserve ratio**, set by the Central Bank, is calculated as a proportion of customer deposits and determines how much commercial banks must set aside in storage rather than lend out. Let's assume, for the purpose of this simulation that there is a mandated reserve ratio of 10%.

- For example, if a bank receives $100 and has a reserve requirement of 10%, it can lend out $90. If this loan is deposited in another bank, that bank can lend out $81, and so on.

- This creates a **multiplier effect**, increasing the overall **money supply**. This can stimulate the economy, but can also cause inflation if the money supply grows too quickly.

- To find out how much money is created with a specific percent ratio, you can use a formula.

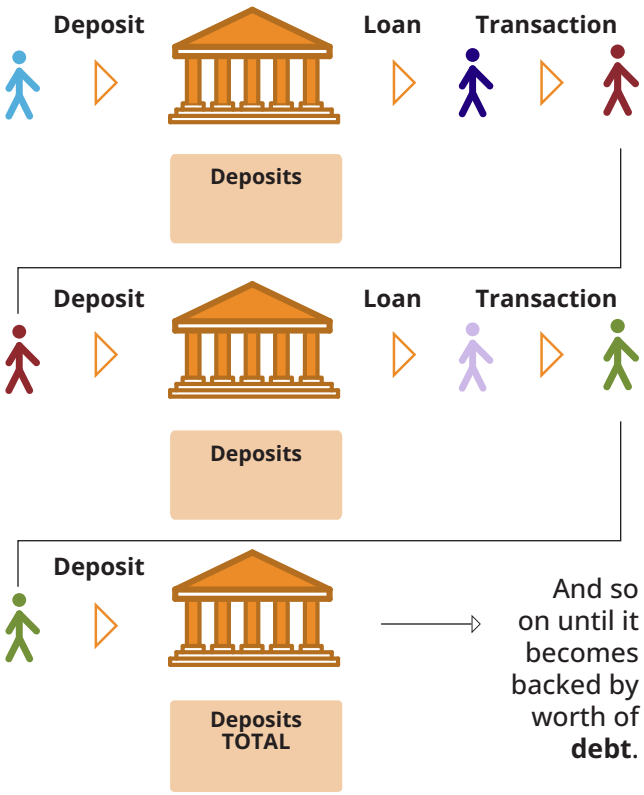
- To use the formula, you first need to know the total amount of money in the economy. This is all the money that is being used to buy and sell goods and services. Then, you need to know the reserve ratio, which is the percentage of money that a bank must keep on hand and not lend out.

- In summary, when a bank lends out money, it creates new money that didn't exist before, and this increases the total amount of money in the economy.

- Generally, countries with volatile economies or high levels of inflation have high reserve ratios to help mitigate risks and to stabilize the financial system.

We need the following volunteers:

A = Depositor (Lottery Winner) (Light Blue)
B = Bank Cashier (Bank)
C = Debtor #1 (Dark Blue)
D = Property Owner/Depositor (Red)
E = Debtor #2 (Light Purple)
F = Art Gallery Owner/ Depositor (Green)



Uncovering the Dark Side of Fiat

A just won $100,000 from the lottery and goes to a newly-opened bank to deposit it. The bank, **B**, has a 10% ratio requirement. How much is **B** required to keep in its vault? _____.

The next morning, **C** enters the bank and asks for a loan. How much can the bank lend out?_____.

C borrows the maximum amount because he wants to put a down payment on a house.

C endorses the check and hands it over to **D**. **D** then goes to the bank and deposits the check. How much did **D** deposit? _____. What are the total deposits recorded in the bank at the moment?_____.

E enters the bank and asks for a huge loan. The bank says that they can lend them at most _____.

E walks out of the bank with the money and goes to buy a piece of art from **F**. After a back-and-forth negotiation, the art piece is sold exactly for what **E** borrowed from the bank. **E** pays **F**.

F deposits the money in the bank. What are the total deposits recorded at the moment?_____.

Name	Deposit	Loan Amount	Reserve Amount
A			
C			
D			
E			
F			

So, how much money is actually created with those 100,000 USD if the money continues to circulate throughout the economy?

_____

_____

When the reserve ratio is high, banks have to keep more money in their safe and can lend out less. This can make it harder for people and businesses to borrow money and can slow down the economy. When the reserve ratio is low, banks have to keep less money in their safes and can lend out more. This can make it easier for people and businesses to borrow money and can make the economy grow faster.

So can we figure out the answer for our class exercise where the reserve is 10%?
(Remember to convert 10% to decimal form 10%=0.1)

Just out of curiosity, how much money would be created in an economy if its reserve ratio was lowered to 1%? (Make sure you divide $100,000/0.01). Surprised?

- As of 2020, the Federal Reserve (the Central Bank of the USA) **reduced reserve requirement ratios to zero percent** in order to stimulate the economy.